Risk-Gated Mechanization: Severity, Verifiability, and Decisions Invariant to the Executor's Error Law

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Abstract

When should a step of agentic work be replaced by a deterministic mechanism? Earlier papers in this series answered with escape probabilities and costs, and treated every escaped fault as equally bad. We add severity and formalize a two-way principle, which reads informally as: mechanize a high-severity step whose correctness can be verified, and do not mechanize one whose correctness cannot. Read literally, the principle overreaches; we give its exact conditions. Severity sets a risk budget; it does not decide alone. An option is admissible when its worst-case loss over the models compatible with the evidence fits the budget. Mechanization is mandatory when closure is admissible and every other available option is not, and forbidden when a compatible closed model exceeds the budget; when no option is admissible, the step needs more evidence, an exogenous gate, or a redesign. Verification is a sufficient way to make closure admissible, and an evidence-compatible closed model whose expected loss exceeds the budget, such as an indistinguishable unsafe twin, forbids it. The bounds that decide are an executor's conditional escape floor and ceiling and a mechanism's certified ceiling. They hold uniformly over every error law, dependence structure, adaptive routing, and mixture of executors that satisfies them; a stochastic-dominance theorem extends this invariance to every nondecreasing loss. When evidence-indistinguishable worlds prefer opposite actions, every evidence-based rule has worst-case regret that grows linearly with severity. A martingale budget gives a scheduler an anytime guarantee under adaptive routing. Persistence enters through convex order: within conditionally independent processes, an uncertain mechanism fault mass is worse than fresh errors of the same mean for every convex loss, by the convexity of Bernstein polynomials, while the probability of any escape moves the other way. Production mixing averages floors; conjunctive review intersects blind sets, and one reviewer dominates another for every producer exactly when its blind set is contained in the other's. With linear loss and constant severity, the results reduce to the earlier mechanization criterion. The decision-theoretic and probabilistic tools are classical; the contribution is their application to the mechanization decision. These are mathematical models, not measurements.

1 Introduction

Part I of this series models gated computation and shows that a gate family with a common blind set leaves every executor with a verification floor [1]. Part II shows that gates generated within one family inherit its blind set, and that exogenous or exact evidence is needed to lower the

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floor [2]. Part III treats escape, cost, and correlated escape [3]. Part IV studies progressive semantic mechanization: moving work from executor calls to deterministic mechanisms [4]. It shows that closure helps only when the mechanism's escape-plus-fallback risk is below the open path's risk, and that evidence cannot separate *twins*, mechanisms with the same observation law. Part V removes the constant-fault-rate assumption and works with conditional risks [5].

None of these papers weighs how bad a fault is. This paper adds severity through the following principle.

> **Risk-gated mechanization.** Consequence-weighted risk is the main gate for mechanization. If a step's correctness is verifiable and its severity is high, the step must be mechanized. If its correctness is not verifiable and its severity is high, it must not be mechanized; it stays open behind a human or exact-evidence gate. A fully determined step is the simplest verifiable case.

Read literally, the principle is a heuristic. Verifiable closure is not mandatory if another option, such as an exact added gate, also fits the budget; and failure to verify does not forbid closure if every compatible closed model is safe. Theorem 7 gives the exact conditions.

A second aim shapes the results. Executors differ: a person, and several language models at different effort levels, have different and unknown error laws, and work is routed among them. A scheduler's decision rule should not depend on knowing those laws. We therefore look for statements that use only bounds, and that hold uniformly over every error law satisfying them. *Invariance* in this paper always means this uniformity. Without any bound, no positive floor and no mandate to mechanize can exist, because a fault-free executor is among the possible laws.

Contributions.

1. Classes of executor processes defined by full-history conditional escape bounds, closed under mixtures and adaptive routing, with the classical binomial domination of such processes [16] (Lemma 3, Theorem 4).
2. Robust admissibility under a severity-derived budget, which gives "must" and "must not" a precise meaning, including the case where no option is admissible (Theorem 7).
3. The regret price of the second arm: for observational twins, Le Cam's two-point bound and minimax-regret allocation give a worst-case regret linear in severity (Theorem 10).
4. Expected-loss thresholds for both arms, with the actual-risk, ex-ante, and worst-case readings kept apart, and the classical sharp convex bounds from the mean applied to mean-only certificates (Theorem 11, Proposition 12).
5. Persistence versus fresh errors as a convex-order comparison of mixed binomial counts (Theorem 14).
6. An anytime consequence budget for an adaptive scheduler, from the standard supermartingale construction (Theorem 17).
7. Production mixing, reviewer conjunction, and the statewise-dominance condition for universal reviewer superiority, in the gated form of design-diversity models (Proposition 18).
8. Structural severity as a weighted reach, with a damage envelope and an ancestry bound (Proposition 20).
9. A reduction to Part IV's criterion (Proposition 21).

The probabilistic and decision-theoretic tools are classical: coupling and domination by independent variables, stochastic and convex orders, Bernstein polynomials, total variation and two-point bounds, minimax risk and minimax regret, nonnegative supermartingales, and positive association. The contribution is their application to the mechanization decision and the invariance they give. Section 14 gives the sources.

2 Setting

Units, escapes, and losses. A step of work is executed on n units, where n is fixed unless stated otherwise. Let $J_i \in \{0,1\}$ indicate that unit i emits an escaped fault, and $N_n = \sum_{i \le n} J_i$. Let $\mathcal{F}_{i-1}$ be the full history before unit i: earlier outputs and escapes, verdicts, routing choices, and internal states. A *loss function* is a nondecreasing $\varphi : \{0, \dots, n\} \to [0, \infty)$ with $\varphi(0) = 0$. Examples are the expected count $\varphi(k) = k$, the any-escape indicator $\varphi(k) = \mathbf{1}\{k \ge 1\}$ of Part V's success metric, and convex losses that penalize concentrated damage. A step has severity $S \ge 0$ (Section 10); its loss is $S\varphi(N_n)$.

Executor classes.

Definition 1 (Conditional escape class)**.** For $0 \le \ell \le u \le 1$, the class $\mathcal{P}(\ell, u)$ contains every escape process with

$$\ell \le \mathbb{P}(J_i = 1 \mid \mathcal{F}_{i-1}) \le u \quad \text{almost surely, for every } i \le n.$$

The lower bound is a *floor*. Under Part I's model, with first-attempt badness $p(h)$ and a stealth share $\sigma(h)$ that passes every gate at once, the floor at history h is at least $p(h)\sigma(h)$ [1, 5]. The upper bound is a *ceiling*; independent majority gating drives it down, but not below the floor [1, 5]. The class says nothing about arrival laws, dependence, retries, or the shape of the error law. The bounds must hold on the full history, including histories after earlier escapes; a floor on correct prefixes only is weaker (Example 5). Under routing, an executor's bound must hold on the histories and tasks actually routed to it; an unconditional benchmark does not supply it.

Mechanisms and evidence. A mechanism M maps inputs to outputs. Its fault mass $\pi(M)$ is the probability that an input from the target law is mapped to a faulty output. With a retained gate and a fallback path, its per-unit escape is $d(M) = e(M) + b(M)f$, as in Parts IV and V. We write $a(M)$ and $A(M)$ for lower and upper bounds on its conditional per-unit escape that hold at every history. When deployment is ungated, or the retained gate accepts every faulty output, inputs are independent draws from the target law, and there is no fallback, both equal $\pi(M)$; repeated or adaptively selected inputs need not satisfy this. A retained gate can make escape smaller than fault mass: an exact fail-closed gate on an always-faulty mechanism gives escape 0. With a gate or fallback, the escape bounds come from the gate and fallback model. A gate that only suppresses outputs keeps escape at most the fault mass, so a fault-mass bound remains an escape ceiling; a fallback can add escapes, and then validation of raw fault mass does not by itself certify the closed path's escape.

A promotion rule sees evidence: the full record of an evidence experiment, including any adaptive or sequential choices, and auxiliary randomness. Two mechanisms are *twins* for that experiment if they induce the same law for its full record. Twin status is relative to the experiment: equality of one-sample record laws does not imply equality for two samples (two independent fair bits and two identical fair bits agree on one bit and differ on two). Independent repetitions of an identical

per-sample record law preserve twin status at every sample size. Let $\mathcal{C}$ be the mechanisms compatible with the stated assumptions. Two twin floors are needed, one for raw fault mass and one for closed-path escape:

$$\tau_\pi(M) = \sup\{\pi(M') : M' \in \mathcal{C} \text{ a twin of } M\}, \qquad \tau(M) = \sup\{a(M') : M' \in \mathcal{C} \text{ a twin of } M\}.$$

Part IV shows that a valid upper bound on fault mass, computed by a fixed rule from the record, is at least $\tau_\pi(M)$ with high probability [4, 5]; the same argument shows that a valid upper bound on closed-path escape is at least $\tau(M)$ with high probability. Each certificate must be compared with its own floor: a fault-free mechanism whose retained gate always rejects and whose fallback always emits a fault has $\pi(M) = 0$ but $a(M) = 1$. Certifying escape needs validation that covers the whole closed path, including the gate and fallback. Twin floors limit what evidence can certify; they are not lower bounds on actual risk: a fault-free mechanism can have a faulty twin.

Definition 2 (Verifiable)**.** Fix a level U in advance. A step is *verifiable at level U with confidence* $1 - \eta$ if an available build-and-evidence experiment promotes a mechanism with $\mathbb{P}(A(M) > U \mid \text{promoted}) \le \eta$.

The level is fixed before the experiment; a data-dependent bound is not treated as having posterior error η. The guarantee is about the randomized build-and-evidence experiment, averaged over promoted builds; it is not a guarantee for every build compatible with the evidence, even with $\eta = 0$. Robust admissibility of a deployed build needs the uniform premise of Theorem 7(iii). A fully determined step, whose decision tree is known, closed, and checked exactly on every branch, is verifiable at level $U = 0$ with $\eta = 0$. Zero-failure validation on m independent target samples bounds the joint probability of promotion with fault mass above U by $(1 - U)^m$, the classical zero-failure demonstration bound [23, 24]; the conditional level divides this by the promotion probability [5].

Gates and blind sets. A producer e emits a bad unit with probability p_e; given badness, its fault has law μ_e. A gate accepts every fault in its blind set. A blind set is *complete* if it is exactly the set of accepted faults, and *guaranteed* if it is only known to be accepted. A conjunctive stack accepts a fault only if every member accepts it [2].

3 Invariance lemmas

Lemma 3 (Mixtures and routing stay in the class)**.** *Let K be a component index, possibly chosen adaptively from the history, and suppose $\ell \le \mathbb{P}(J_i = 1 \mid \mathcal{F}_{i-1}, K) \le u$ almost surely for every i. Then the mixed process is in $\mathcal{P}(\ell, u)$. In particular, adaptive routing among executors whose conditional escape lies in $[\ell_e, u_e]$ on the routed histories yields a process in $\mathcal{P}(\min_e \ell_e, \max_e u_e)$; with routing weights $w_{i,e}$, the conditional escape is $\sum_e w_{i,e} q_{i,e} \ge \sum_e w_{i,e} \ell_e$.*

Proof. By the tower property, $\mathbb{P}(J_i = 1 \mid \mathcal{F}_{i-1}) = \mathbb{E}[\mathbb{P}(J_i = 1 \mid \mathcal{F}_{i-1}, K) \mid \mathcal{F}_{i-1}]$, an average of values in $[\ell, u]$. □

Theorem 4 (Binomial dominance)**.** *For every process in $\mathcal{P}(\ell, u)$,*

$$\mathrm{Bin}(n, \ell) \ \le_{\mathrm{st}} \ N_n \ \le_{\mathrm{st}} \ \mathrm{Bin}(n, u).$$

Hence $\mathbb{E}\varphi(\mathrm{Bin}(n, \ell)) \le \mathbb{E}\varphi(N_n) \le \mathbb{E}\varphi(\mathrm{Bin}(n, u))$ for every loss function φ. For the any-escape indicator, a floor on correct prefixes alone gives Part V's survival bound $\mathbb{P}(N_n = 0) \le (1 - \ell)^n$.

Proof. This is the classical domination of a binary sequence with bounded conditional probabilities by independent variables [16, Lemma 1.8.7], stated there for conditioning on past indicators, which the tower property gives from bounds given $\mathcal{F}_{i-1}$. We include the coupling. On an enlarged probability space, take independent uniform variables $V_1, \dots, V_n$ independent of everything else. Construct the process sequentially: given the history, set $J_i = \mathbf{1}\{V_i \le \mathbb{P}(J_i = 1 \mid \mathcal{F}_{i-1})\}$ and draw the remaining history variables from their conditional law given J_i. This reproduces the joint law. Put $L_i = \mathbf{1}\{V_i \le \ell\}$ and $H_i = \mathbf{1}\{V_i \le u\}$. Then $L_i \le J_i \le H_i$ pointwise, with $\sum L_i \sim \mathrm{Bin}(n, \ell)$ and $\sum H_i \sim \mathrm{Bin}(n, u)$. Pointwise order gives stochastic order, and φ is nondecreasing. The survival bound is Part V's conditional floor. □

Example 5 (A correct-prefix floor does not order counts)**.** Let J_1 be Bernoulli(1/2) and $J_2 = 1 - J_1$. The conditional risks on correct prefixes are 1/2 and 1, both at least 1/2, but $N_2 = 1$ always, which is not stochastically above $\mathrm{Bin}(2, 1/2)$: the binomial gives $\mathbb{P}(N \ge 2) = 1/4$. Likewise a marginal ceiling does not give a binomial ceiling: $J_1 = J_2 = Z$ with Z Bernoulli(0.1) has $\mathbb{P}(N_2 = 2) = 0.1 > 0.01$.

Remark 6. Any statement that uses only $\mathbb{E}\varphi(\mathrm{Bin}(n, \ell))$ and $\mathbb{E}\varphi(\mathrm{Bin}(n, u))$ therefore holds uniformly over arrival laws, dependence, retry laws, and every mixture of executors within the bounds. The map $t \mapsto \mathbb{E}\varphi(\mathrm{Bin}(n, t))$ is a nondecreasing polynomial, hence continuous. An outcome-dependent number of units needs the stopping-time form of Section 8.

4 Robust admissibility under a risk budget

Severity enters as a budget. Fix a tolerated expected loss $K > 0$ for a step of severity $S > 0$. For an option a, let $\mathcal{C}_a$ be the set of models compatible with the available evidence and assumptions, and define its worst-case expected loss

$$r_a^+ = \sup_{P \in \mathcal{C}_a} \mathbb{E}_P \varphi(N_a).$$

This is the maximum risk of Wald's minimax criterion [17], taken over the set of compatible models, as in multiple-prior decision theory and partial identification [18, 19]. Option a is *robustly admissible*[1] if $S r_a^+ \le K$. Under linear loss, r_a^+/n is the worst-case per-unit escape, and admissibility reads $r_a^+/n \le r^*$ with the *risk budget* $r^* = K/(Sn)$. High severity means a small budget. For closure we distinguish two readings. A *deployed build* is robustly admissible if the worst case over builds compatible with the evidence fits the budget. A *build-and-evidence experiment* is admissible ex ante if the worst case, over experiments satisfying Definition 2, of the expected loss averaged over promoted builds fits the budget.

Theorem 7 (Admissibility; the two arms)**.** *Consider the options* open *(executors with a gate stack),* gated *(open with an added gate H from an available set), and* closed *(a promoted mechanism).*

(i) Invariance. *Replacing $\mathcal{C}_a$ by its convex hull does not change r_a^+, so whole-process mixtures of compatible models leave admissibility unchanged. If $\mathcal{C}_a$ is a bound class $\mathcal{P}(\ell, u)$, it is also closed under adaptive routing (Lemma 3). A compatible set given by specific executor laws need not be: routing can produce a process outside its convex hull.*

[1] Here *admissible* means that the option fits the budget. In statistical decision theory the word means that a rule is not dominated [17]; the two notions are unrelated.

(ii) Open path. *If the open path's compatible set is $\mathcal{P}(\ell, u)$, then r^{+}_{open} lies between $\mathbb{E}\varphi(\text{Bin}(n, \ell))$ and $\mathbb{E}\varphi(\text{Bin}(n, u))$. If $S\mathbb{E}\varphi(\text{Bin}(n, \ell)) > K$, the open path is inadmissible for every error law, dependence structure, and routing within the bounds. A gate generated within the producing family leaves the floor unchanged under Part II's inheritance assumptions.*

(iii) Closure, exact certificate. *If every build compatible with the evidence has its closed units in $\mathcal{P}(0, U)$, the deployed build is robustly admissible whenever $S\mathbb{E}\varphi(\text{Bin}(n, U)) \le K$. A fully determined step checked exactly on every branch gives this with $U = 0$.*

(iv) Closure, statistical certificate. *If the step is verifiable at level U with confidence $1 - \eta$, and given promotion the closed units are in $\mathcal{P}(0, A(M))$, then the expected loss of a promoted build, averaged over the experiment, is at most $(1 - \eta)\mathbb{E}\varphi(\text{Bin}(n, U)) + \eta\varphi(n)$. This gives ex-ante admissibility of the experiment, not robust admissibility of the deployed build: an always-promoted build that is fault-free with probability 0.99 and always faulty otherwise has $U = 0$ and $\eta = 0.01$, average risk 0.01, and worst-case risk 1.*

(v) Closure forbidden. *If some compatible twin M' of M has closed units in $\mathcal{P}(a(M'), 1)$ and $S\mathbb{E}\varphi(\text{Bin}(n, a(M'))) > K$, closure is not robustly admissible. In particular this holds when $S\mathbb{E}\varphi(\text{Bin}(n, \tau(M))) > K$. If the twin relation holds for the whole evidence experiment at every sample size, no amount of that evidence changes the conclusion.*

(vi) Gated open path. *With complete blind sets, adding a gate H gives producer e the first-attempt floor $p_e\mu_e(\Sigma \cap \Sigma_H)$. If Σ_H is only guaranteed, this is a lower bound; lowering the ceiling needs knowledge of what H rejects. Exact evidence on a scope removes escape there if it is sound, bound to the executed artifact or action, and enforced fail-closed. A human gate has no special status: its effect is its acceptance set.*

Consequently, closure is mandatory *among the available options when it is admissible in the chosen reading and the open path and every available gated option are inadmissible. Closure is* forbidden *when (v) holds. If no option is admissible, the step needs more distinguishing evidence, an additional exogenous gate, a lower severity through redesign, or must not be performed.*

Proof. (i) A supremum of expectations over a set equals the supremum over its convex hull, because an expectation is linear in the model. Routing closure of $\mathcal{P}(\ell, u)$ is Lemma 3. For the last claim, let executor A always produce escapes $(1, 0)$ on two units and executor B always $(0, 1)$: every whole-process mixture has count 1, while routing unit 1 to A and unit 2 to B gives 2. (ii) Theorem 4 gives both bounds, and the lower bound applies to every compatible law. A generated gate keeps the family's blind set (Part II), so the first-attempt stealth share and the floor are unchanged. (iii) By Theorem 4, every compatible build has expected loss at most $\mathbb{E}\varphi(\text{Bin}(n, U))$. (iv) On the event $A(M) \le U$, of conditional probability at least $1 - \eta$, the closed count is stochastically below $\text{Bin}(n, U)$; otherwise the loss is at most $\varphi(n)$. The example has the stated values. (v) The twin is compatible and its expected loss is at least $\mathbb{E}\varphi(\text{Bin}(n, a(M')))$ by Theorem 4. The supremum over compatible twins of that quantity is $\mathbb{E}\varphi(\text{Bin}(n, \tau(M)))$, by monotonicity and continuity. If twins stay twins at every sample size, the compatible set never excludes M'. (vi) A fault passes the conjunction only if every member accepts it; with complete sets the passing set is exactly $\Sigma \cap \Sigma_H$, and with guaranteed sets it contains it. Sound, bound, fail-closed exact evidence accepts no fault on its scope. □

Corollary 8 (Below the floor, inherited gates are not enough)**.** *If $S\mathbb{E}\varphi(\text{Bin}(n, \ell_e)) > K$ for every executor e, no open path whose gates all come from the producing family is admissible, under any routing. An admissible option, if one exists, is a closure with $Sr^{+}_{\text{closed}} \le K$, or an open path*

gated outside the family. Parts (iii), (iv), and (vi) give sufficient ways to establish these; they are not necessary. For example, with $n = 2$, linear loss, $S = 1$, $K = 0.05$, and an open escape of 0.1 per unit, a closed process with $J_1 = J_2 = Z$ and $Z \sim$ Bernoulli(0.01) is robustly admissible ($\mathbb{E}N_2 = 0.02$), although its uniform conditional ceiling is 1, so neither (iii) nor (iv) certifies it.

Proof. By Theorem 7(ii) and Lemma 3, every routing of an open path with inherited gates has expected loss at least $S\mathbb{E}\varphi(\mathrm{Bin}(n, \min_e \ell_e)) > K$. The example has $\mathbb{E}N_2 = 2 \cdot 0.01$ and $\mathbb{P}(J_2 = 1 \mid J_1 = 1) = 1$. □

Remark 9. The theorem gives the principle precise sufficient conditions, not unconditional implications. *Verifiable at a level within budget* is a sufficient way to make closure admissible; with a statistical certificate the admissibility is ex ante. *High severity* means a budget below the executor floors. A step that is not verifiable at a chosen level is not thereby forbidden: closure is admissible when every compatible closed model fits the budget, whether or not twins are separated. A safe twin set is not enough, because compatible models need not be twins: a safe M_0 with evidence Bernoulli$(1/2)$ and an unsafe M_1 with evidence Bernoulli$(1/4)$ both remain compatible after observing 0. Closure is forbidden when some compatible closed model violates the budget; a violating twin is one way this happens. Severity alone decides nothing: it sets the budget, and the comparison with the floors, the certified level, and the compatible twins decides. Part (v) concerns what evidence permits, not actual risk; Theorem 10 prices the gap.

5 Decisions on twins: severity-scaled regret

Keeping a step open because closure cannot be certified may be wrong in a world where the mechanism is in fact fault-free. When evidence-indistinguishable worlds prefer opposite actions, evidence cannot remove this dilemma. Unverifiability alone does not create it: if indistinguishable mechanisms of fault mass 0.8 and 0.9 face an open path of escape 0.1, always keeping the step open has zero regret.

Theorem 10 (Twin regret)**.** *Consider two worlds whose evidence records have laws Q_0, Q_1 with $d_{\mathrm{TV}}(Q_0, Q_1) \le \varepsilon$, and suppose that in each world the deployment losses are independent of the evidence record. In world 0, closure beats the open path by expected total loss $\alpha S > 0$; in world 1, the open path beats closure by $\beta S > 0$. Every decision rule based on the evidence, possibly randomized, has*

$$\max\{R_0, R_1\} \;\ge\; (1 - \varepsilon)\,\frac{\alpha\beta}{\alpha + \beta}\,S,$$

where R_j is the expected regret in world j, measured against the better action's expected loss. For exact twins, $\varepsilon = 0$ and the bound is attained by choosing closure with probability $\alpha/(\alpha + \beta)$.

Proof. Let p_j be the probability of choosing closure in world j. The data-processing property of total variation gives $|p_0 - p_1| \le \varepsilon$. Because losses are independent of the evidence, the expected loss of a rule is p_j times the closed loss plus $1 - p_j$ times the open loss, so the regrets are $R_0 = \alpha S(1 - p_0)$ and $R_1 = \beta S p_1$, so $R_0/(\alpha S) + R_1/(\beta S) = 1 - p_0 + p_1 \ge 1 - \varepsilon$. Since $\max\{R_0, R_1\}(1/(\alpha S) + 1/(\beta S)) \ge R_0/(\alpha S) + R_1/(\beta S)$, the bound follows. With $\varepsilon = 0$ and $p_0 = p_1 = \alpha/(\alpha + \beta)$, both regrets equal $\alpha\beta S/(\alpha + \beta)$. □

The theorem combines two classical results. The inequality $1 - p_0 + p_1 \ge 1 - \varepsilon$ is Le Cam's two-point argument [20]. The exact-twin rule and its regret are the two-state case of minimax-regret treatment allocation [21, 22], with αS and βS as the welfare gaps. What is added here is

their reading as the cost of the second arm. The independence assumption matters: if the evidence predicts deployment losses, a rule can exploit it. With a fair evidence bit X and losses (closed, open) equal to $(0,2)$ or $(1,0)$ in world 0 and $(0,1)$ or $(2,0)$ in world 1 according to X, the unconditional preferences reverse, yet closing exactly when $X = 0$ has zero loss in both worlds. Under the assumption, the unavoidable worst-case regret grows linearly with severity. A richer scheduler cannot resolve it with evidence that remains indistinguishable; when the transcript includes an adaptive build-and-test process, the twin relation is taken over that whole transcript. Only evidence that separates the worlds, or a change of available actions such as an exogenous gate, reduces it. This is the precise cost behind the second arm of the principle.

6 Expected loss: thresholds for both arms

Admissibility uses a hard budget. A softer formulation compares expected severity-weighted loss plus cost. Let C_{op} and C_{cl} be the expected costs of the open and closed paths over the n units, including build, validation, upkeep, gates, retries, and fallback.

Theorem 11 (Two arms under expected loss)**.** *Let the open path be in $\mathcal{P}(\ell, u)$ and let φ be a loss function.*

(a) Mandatory closure (ex ante). *Under the assumptions of Theorem 7(iv), put $\Delta_1 = \mathbb{E}\varphi(\mathrm{Bin}(n,\ell)) - (1-\eta)\mathbb{E}\varphi(\mathrm{Bin}(n,U)) - \eta\varphi(n)$. If $\Delta_1 > 0$, closure has lower ex-ante expected loss plus cost than the open path whenever $S > S_1^* = (C_{\mathrm{cl}} - C_{\mathrm{op}})_+/\Delta_1$. If instead the uniform premise of Theorem 7(iii) holds, take $\eta = 0$; the comparison then holds for every compatible model, not only in expectation. A statistical certificate with $\eta = 0$ alone does not give this.*

(b) Forbidden closure (worst case). *Under the assumptions of Theorem 7(v), put $\Delta_2 = \mathbb{E}\varphi(\mathrm{Bin}(n,\tau(M))) - \mathbb{E}\varphi(\mathrm{Bin}(n,u))$. If $\Delta_2 > 0$, the worst-case expected loss plus cost of closure over the compatible twins exceeds that of the open path whenever $S > S_2^* = (C_{\mathrm{op}} - C_{\mathrm{cl}})_+/\Delta_2$.*

Both thresholds depend on the executors only through ℓ and u. A threshold exists only when the corresponding gap is positive: a valid but loose certificate with $U \geq \ell$ gives no mandate.

Proof. (a) The open path's expected loss is at least $S\mathbb{E}\varphi(\mathrm{Bin}(n,\ell))$ by Theorem 4, and the closed path's is at most $S[(1-\eta)\mathbb{E}\varphi(\mathrm{Bin}(n,U)) + \eta\varphi(n)]$ by Theorem 7(iv). Compare the totals. (b) The worst case over compatible twins of the closed loss is at least $S\mathbb{E}\varphi(\mathrm{Bin}(n,\tau(M)))$, and the open loss is at most $S\mathbb{E}\varphi(\mathrm{Bin}(n,u))$. Compare the totals. □

Part (b) is a worst-case preference. It does not say that the open path is better in every compatible world: with twins of fault mass 0 and 0.9, an open path of escape 0.1, equal costs, and linear loss, (b) prefers the open path while the fault-free twin favours closure. Theorem 10 prices that gap. For a fully determined step, $U = 0$ and $\eta = 0$, so $\Delta_1 = \mathbb{E}\varphi(\mathrm{Bin}(n,\ell)) > 0$ whenever $\ell > 0$ and $\varphi(1) > 0$, and S_1^* is finite for every cost difference.

Mean-only certificates. Validation may certify average fault mass without certifying a conditional ceiling.

Proposition 12 (Sharp convex bounds from the mean)**.** *Let $N \in \{0,\dots,n\}$ with $\mathbb{E}N = m < n$, and let φ be convex. With $k = \lfloor m \rfloor$ and $\alpha = m - k$,*

$$(1-\alpha)\varphi(k) + \alpha\varphi(k+1) \;\leq\; \mathbb{E}\varphi(N) \;\leq\; \Big(1 - \frac{m}{n}\Big)\varphi(0) + \frac{m}{n}\varphi(n),$$

and both bounds are attained. If φ is a nondecreasing convex loss and $\mathbb{E}N \le nU$, then $\mathbb{E}\varphi(N) \le U\varphi(n)$, and closure has lower expected loss plus cost than an open path in $\mathcal{P}(\ell, u)$ whenever $S[\mathbb{E}\varphi(\mathrm{Bin}(n,\ell)) - U\varphi(n)] > C_{\mathrm{cl}} - C_{\mathrm{op}}$.

Proof. The lower bound is Jensen's inequality for the piecewise-linear interpolation of φ, attained by a law on $\{k, k+1\}$. The upper bound is the chord of φ between 0 and n, attained by a law on $\{0, n\}$. Each extremal law is the count of an exchangeable Bernoulli sequence: choose a uniformly random subset of the drawn size. For the last claim, $\varphi(0) = 0$ and the chord bound is nondecreasing in m. Both bounds are classical: the upper one is the Edmundson–Madansky bound [25], and both are the discrete convex extrema with a given mean [26]. □

The gap between $\mathbb{E}\varphi(\mathrm{Bin}(n, U))$ and $U\varphi(n)$ measures what is lost when validation cannot certify how faults cluster.

7 Persistence and convex order

Linear loss sees only the mean escape. A convex loss also sees how escapes cluster. Part V showed that persistence, a shared defect or a latent regime, changes the damage distribution at equal mean [5]. Convex order makes this exact. For integer random variables write $X \le_{\mathrm{cx}} Y$ if $\mathbb{E}g(X) \le \mathbb{E}g(Y)$ for every convex g; this forces equal means. For a function f on $\{0, \dots, n\}$, the Bernstein polynomial is $B_n f(\theta) = \sum_{k=0}^{n} f(k)\binom{n}{k}\theta^k(1-\theta)^{n-k} = \mathbb{E}f(\mathrm{Bin}(n,\theta))$.

Lemma 13 (Bernstein polynomials preserve convexity)**.** *For $n \ge 2$,*

$$(B_n f)''(\theta) = n(n-1)\sum_{k=0}^{n-2} \Delta^2 f(k)\binom{n-2}{k}\theta^k(1-\theta)^{n-2-k}, \qquad \Delta^2 f(k) = f(k+2) - 2f(k+1) + f(k).$$

Hence $B_n f$ is convex on $[0,1]$ if f is convex, and strictly convex if every $\Delta^2 f(k) > 0$.

Proof. Differentiate twice and collect terms; this is the classical identity behind convexity preservation [6]. □

Theorem 14 (Persistence is convex spread)**.** *Let $J_1, \dots, J_n$ be conditionally independent Bernoulli(Θ) given a random $\Theta \in [0,1]$, and let J' be the same with Θ'. If $\Theta \le_{\mathrm{cx}} \Theta'$, then $N_n \le_{\mathrm{cx}} N'_n$. At a given mean escape $\bar{\theta}$:*

- *(a) fresh independent errors, $\Theta \equiv \bar{\theta}$, are convex-minimal among such processes;*
- *(b) a closed mechanism drawn from a build process, deployed ungated with independent inputs so that all units share $\Theta = \pi(M)$, has $\mathrm{Bin}(n, \bar{\theta}) \le_{\mathrm{cx}} N_n$, strictly for strictly convex losses when $n \ge 2$ and $\pi(M)$ is not almost surely constant;*
- *(c) linear loss cannot distinguish them: $\mathbb{E}N_n = n\bar{\theta}$ in both cases, while $\operatorname{Var} N_n = n\bar{\theta}(1-\bar{\theta}) + n(n-1)\operatorname{Var}\Theta$.*

Proof. For convex g, $\mathbb{E}g(N_n) = \mathbb{E}[B_n g(\Theta)]$, and $B_n g$ is convex by Lemma 13, so $\Theta \le_{\mathrm{cx}} \Theta'$ gives $\mathbb{E}g(N_n) \le \mathbb{E}g(N'_n)$. Part (a): a constant is convex-below every variable with the same mean, by Jensen. Part (b) is (a) with strict Jensen for the strictly convex $B_n g$. Part (c): B_n maps linear functions to linear functions, and the variance follows from the law of total variance. □

The implication from $\Theta \le_{\mathrm{cx}} \Theta'$ to $N_n \le_{\mathrm{cx}} N'_n$ for mixed binomial counts is known [7, 27]; the new part is its reading as persistence against fresh errors. Part (a) concerns mixtures only: independent errors with unequal probabilities and the same mean are convex-below $\mathrm{Bin}(n, \bar{\theta})$ [28].

Remark 15. Three limits apply. First, the comparison is within conditionally independent processes; finite exchangeability is not enough: a uniformly random ordering of $(0, 1)$ is exchangeable with constant count 1, convex-below $\mathrm{Bin}(2, 1/2)$. Second, the mixture over builds expresses uncertainty about which build is deployed; for a single deployed build the count is binomial with its own fault mass. A persistent executor regime drawn once per run (Part V) is also a mixture, so the contrast is persistent versus fresh, not mechanism versus executor. Third, the any-escape indicator is not convex and moves the other way: with $n = 2$ and mean $1/2$, complete persistence gives $\mathbb{P}(N \geq 1) = 1/2$ against $3/4$ for fresh errors, while $\mathbb{P}(N = 2)$ rises from $1/4$ to $1/2$. Persistence trades frequent small damage for rare concentrated damage; the loss shape decides which matters.

Example 16 (Same mean, different risk)**.** Take $n = 10$, mean escape 0.1, and $\varphi(k) = \max(0, k-2)^2$, a loss that starts after two escapes. Fresh errors give $\mathbb{E}\varphi = 0.1179$. A mechanism with fault mass 0.02 or 0.18 with equal probability gives 0.3327. Under $\varphi(k) = k$ both give 1.0000; the variances are 0.900 and 1.476.

8 An anytime budget for a scheduler

A scheduler chooses actions unit by unit, and the number of units may depend on outcomes. Local certificates then need to add up to a lifetime guarantee.

Theorem 17 (Adaptive consequence budget)**.** *At each unit the scheduler chooses, from the history, an action, a weight $s_i \geq 0$, and a conditional escape ceiling u_i with $\mathbb{P}(J_i = 1 \mid \mathcal{F}_{i-1}) \leq u_i$. Let $D_t = \sum_{i \leq t} s_i J_i$ and let T be a stopping time.*

(a) If $\sum_{i \leq T} s_i u_i \leq B$ almost surely, then $\mathbb{E}D_T \leq B$. If s_i bounds the conditional expected damage of an escape at unit i given the history, the expected total damage is at most B.

(b) For every fixed $\lambda > 0$ and $\delta \in (0, 1)$,

$$\mathbb{P}\Big(\exists t : \ \lambda D_t - \sum_{i \leq t} \log\left[1 + u_i(e^{\lambda s_i} - 1)\right] \geq \log(1/\delta)\Big) \leq \delta.$$

In (b), D_t bounds the actual damage pathwise only if each s_i bounds the damage of unit i pathwise. Both parts hold under adaptive routing and every mixture of executors whose ceilings hold on the routed histories.

Proof. (a) $\mathbb{E}D_T = \mathbb{E}\sum_{i \leq T} s_i \mathbb{P}(J_i = 1 \mid \mathcal{F}_{i-1}) \leq \mathbb{E}\sum_{i \leq T} s_i u_i \leq B$, by optional stopping for the nonnegative increments with predictable weights; monotone convergence handles unbounded T. For the damage claim, the expected damage of unit i given the history is at most $\mathbb{P}(J_i = 1 \mid \mathcal{F}_{i-1}) s_i$. (b) $\mathbb{E}[e^{\lambda s_i J_i} \mid \mathcal{F}_{i-1}] = 1 + \mathbb{P}(J_i = 1 \mid \mathcal{F}_{i-1})(e^{\lambda s_i} - 1) \leq 1 + u_i(e^{\lambda s_i} - 1)$, so $M_t = \exp(\lambda D_t - \sum_{i \leq t} \log[1 + u_i(e^{\lambda s_i} - 1)])$ is a nonnegative supermartingale with $M_0 = 1$. Ville's maximal inequality gives the bound [13]. This is the standard supermartingale construction of time-uniform Chernoff bounds [29]. □

For expected damage, s_i can be the structural severity of Section 10, an expected damage. For the anytime tail of actual damage, s_i must be a pathwise bound, such as the total cost of the reach set; with expected severities, part (b) controls the severity-weighted escape sum, a proxy. The theorem turns per-step certificates into a budget that the scheduler can spend: closing a step with a verified mechanism lowers its u_i; adding a gate lowers u_i at a cost; high-severity steps consume the budget fastest.

9 Producers average, reviewers intersect

Proposition 18 (Production mixing and review)**.** *Let producers e have badness p_e and fault laws μ_e, and let gates accept their blind sets at once.*

(a) Routing production with weights w_e gives the first-attempt floor $\sum_e w_e p_e \mu_e(\Sigma) \geq \min_e p_e \mu_e(\Sigma)$; lower bounds average as lower bounds.

(b) Conjunctive review by $r_1, \ldots, r_k$ gives producer e the escape $p_e \mu_e(\bigcap_j \Sigma_{r_j})$ if the blind sets are complete, and at least that if they are only guaranteed.

(c) Universal superiority. *Let costs be equal, let blind sets be complete, and for every fault z let some admissible producer have $p_e > 0$ and $\mu_e = \delta_z$. Reviewer r_1 has escape at most that of r_2 for every producer if and only if $\Sigma_{r_1} \subseteq \Sigma_{r_2}$. More generally, let a unit's state z range over a fault-free state z_0 and the fault states, with residual damages $h_j \geq 0$, $h_j(z_0) = 0$, and per-unit costs c_j, so that design j's per-unit total is $c_j + S\,\mathbb{E}h_j(Z)$ (for producer e, $\mathbb{E}h_j(Z) = p_e \int h_j \, d\mu_e$). If every point-mass state law is admissible, including the fault-free state, design 1 is at least as good for every admissible state law if and only if $c_1 + Sh_1(z) \leq c_2 + Sh_2(z)$ for every state z, which requires $c_1 \leq c_2$. With a smaller admissible class only sufficiency holds: if every producer has $p_e = 0.01$, then with $S = 1$, $(c_1, h_1) = (0.005, 0)$ and $(c_2, h_2) = (0, \mathbf{1})$, design 1 costs $0.005 < 0.01$ for every such producer although it is worse at the fault-free state.*

(d) No universal ranking. *Let blind sets be complete, and let producers with positive badness concentrated on $\Sigma_{r_1} \setminus \Sigma_{r_2}$, and others concentrated on $\Sigma_{r_2} \setminus \Sigma_{r_1}$, be admissible. Then each reviewer is strictly better for some producer. Both conditions are needed: with guaranteed sets $\{x\}$ and $\{y\}$ for reviewers that in fact accept both faults, the escapes are equal for every producer; and if every admissible producer has fault law $(0.9, 0.1)$ on (x, y), the reviewer missing only y is always better.*

(e) Coalitions. *Pairwise single-reviewer escapes do not determine a coalition's escape: two reviewers each missing half of a producer's faults can have an intersection of mass 0 or $1/2$.*

Proof. (a) The floor is linear in the fault law. (b) A fault passes only if every reviewer accepts it. (c) Integrate the pointwise inequality for sufficiency; for necessity, take a point mass at a state that violates it. The reviewer case has $h_j = \mathbf{1}_{\Sigma_{r_j}}$ and equal costs. Omitting the fault-free state is an error: with $p_e = 0.01$, $S = 2$, $(c_1, h_1) = (1, 0)$ and $(c_2, h_2) = (0, \mathbf{1})$, the fault states alone favour design 1, while the per-unit totals are 1 and 0.02. (d) Take an admissible producer concentrated on $\Sigma_{r_2} \setminus \Sigma_{r_1}$: with complete sets, r_1 rejects its faults and r_2 accepts them; symmetrically for the other difference. (e) Take faults uniform on $\{1, 2, 3, 4\}$, $\Sigma_{r_1} = \{1, 2\}$, and $\Sigma_{r_2} = \{3, 4\}$ or $\{1, 2\}$. □

Producer floors are governed by bounds that survive every mixture. Reviewer rankings are not invariant to the producer's error law, except under blind-set inclusion. A scheduler therefore needs review data indexed by producer and by reviewer set, not a single quality score per reviewer and not only pairwise data. Parts (b) and (e) are the gated form of design-diversity models of software reliability: the probability that diverse versions fail together is the demand-profile mass of the intersection of their failure sets, and single-version failure probabilities do not determine it [30, 31, 32]. Part (c) is statewise dominance: an option has lower expected loss under every admissible law if and only if its loss is pointwise lower. A reviewer from the producer's own family inherits the producer's blind set (Part II). Errors of language models are substantially correlated in practice, partly through shared architectures and providers, and for larger models also across providers [33]. Part (c) also shows that a gate can be preferable because it blocks the most damaging faults, even if it misses more faults by count.

10 Severity from structure

Definition 19 (Structural severity)**.** Let the work form a directed acyclic dependency graph. Gates may sit on edges. A fault that escapes at node v reaches node w with probability $\rho(v \to w)$, with $\rho(v \to v) = 1$. Reaching w costs $c(w) \geq 0$ unless the damage is undone; let $\iota(w) \in [0, 1]$ be the probability that damage at w is not undone, given that w is reached, and assume this event is independent of how w was reached. The severity of v is the expected damage of one escape at v,

$$S(v) = \sum_w \rho(v \to w)\, c(w)\, \iota(w).$$

It is an expectation, not a bound on realized damage: if an escape reaches a node of cost 100 with probability 0.01, then $S = 1$ while the realized damage can be 100.

The reach set $\{w : \rho(v \to w) > 0\}$ is the blast radius of v; S weights it by cost and irreversibility. With $c\iota \equiv 1$, $S(v)$ is the expected spread of an influence-propagation model [34]; weighting reach by consequence follows the quantitative definition of risk as probability and consequence [35].

Proposition 20 (Properties of structural severity)**.** *(a)* Factorization and its limits. *In general, the expected damage of producer e at v is $R_e(v) = \int h(v, z)\, \nu_e(dz)$, where ν_e is the escaped-fault subprobability measure and $h(v, z)$ the downstream damage of a fault of type z. It factorizes as $\nu_e(all)\, S(v)$ with an executor-independent $S(v)$ exactly when the conditional expected damage of an escape is the same for every executor, for example when propagation, interception, and repair do not depend on the fault type. Without that condition, the* damage envelope *$S_{\max}(v) = \sup_z h(v, z)$ gives $R_e(v) \leq \nu_e(all)\, S_{\max}(v)$ for every executor and every mixture.*

(b) Ancestry bound. *If edges transmit faults independently and $\rho(v \to w)$ is the probability that a transmitting path exists, then $S(v) \geq \rho(v \to w) S(w)$ for every descendant w. If transmission is certain along every path, $S(v) \geq S(w)$. An exact gate between v and w makes $\rho(v \to w)$ small, so faults from v can be far less severe than faults born at w.*

(c) Additivity is an upper bound in expectation. *With several escapes, the sum of severities bounds the expected damage from above when damage is the cost of the union of reached nodes, with equality when the reached sets are disjoint almost surely.*

Proof. (a) The factorization is the definition of a conditional expectation that does not depend on e. The envelope bound integrates $h \leq S_{\max}$. (b) For each node x, the events "v reaches w" and "w reaches x" are increasing in the independent edge variables. Harris's inequality gives $\mathbb{P}(\text{both}) \geq \rho(v \to w)\rho(w \to x)$ [8, 9], and "v reaches x" contains their intersection. Multiply by $c(x)\iota(x) \geq 0$ and sum over x. Certain transmission gives $\rho(v \to w) = 1$. (c) Pathwise, the cost of a union is at most the sum of costs, with equality for disjoint sets; take expectations. □

Under the condition of part (a), risk factorizes into an executor-dependent escape law and a structure-dependent severity, which a scheduler can compute once from the dependency graph and reuse across executors. Otherwise the envelope gives an executor-invariant upper bound. Escape laws, floors, and ceilings must still be estimated per executor.

11 Relation to the earlier papers

Proposition 21 (Reduction to the mechanization criterion)**.** *Let $\varphi(k) = k$, let severity be a constant $S > 0$, and let the expected per-unit escapes of the open and closed paths be o and d.*

Then closure has lower expected loss plus cost exactly when $Sn(o-d) > C_{\mathrm{cl}} - C_{\mathrm{op}}$. With equal costs this is Part IV's criterion $d < o$; with the expected loss included in the per-call saving, it is Part IV's break-even.

Proof. By linearity, expected losses are Sno and Snd, without any independence assumption. Rearrange. □

Substituting bounds for o and d gives a sufficient criterion, not Part IV's exact one. Severity therefore changes the decision only through the cost trade-off, the confidence tail, the gaps between bounds, and nonlinear loss. Theorem 4 with the any-escape indicator recovers Part V's conditional floor. The invariance here is of the same kind as Part V's: conditional risks, not arrival laws, carry the argument.

12 What must be measured

The theorems need four kinds of quantity, and bounds suffice for their decisions.

1. *Executor floors and ceilings.* A floor lies in the blind set of the producing family's gates, so those gates cannot measure it. An exogenous exact audit reveals escapes. A binomial confidence bound from n_a audited outputs, for example Clopper–Pearson [12], estimates a stationary escape parameter under its sampling model; it does not certify a uniform conditional bound under arbitrary dependence or workload shift. Adaptive auditing and repeated looks need time-uniform confidence sequences [14].
2. *Certified mechanism levels.* With m independent representative validation samples and zero failures, the one-sided upper bound is $U = 1 - \eta^{1/m}$ [5]. This is a statistical bound, not proof of zero fault mass, and selection among builds needs simultaneous inference. Evidence that cannot separate twins cannot, with high probability, bring a fault-mass bound below τ_π or an escape bound below τ. A fault-mass certificate certifies closed-path escape when raw faults and escapes are justifiably related: with independent target inputs, a gate that only suppresses outputs, and no fallback, $J_i \le F_i$ for the raw-fault indicator F_i, so $\mathbb{P}(J_i = 1 \mid \mathcal{F}_{i-1}) \le \pi(M)$ and a valid bound on $\pi(M)$ bounds escape with the same confidence. With a fallback or other interaction, validation of the whole closed path, or a known structural bound, is needed.
3. *Severity.* The dependency graph, costs, and irreversibility give $S(v)$ or its envelope.
4. *Review coverage.* Escape must be measured per producer and per reviewer set (Proposition 18).

13 Executable checks

A companion script (`simulate_risk_mechanization.py`) checks the finite claims: Bernstein convexity and its second-derivative identity, binomial dominance by exact enumeration of history-dependent processes, mixture closure, convex order of binomial mixtures, the mean-only bounds, the two-arm thresholds, the twin-certification and regret bounds, the budget supermartingale, the reviewer and coalition examples, and the severity ancestry bound by enumeration. Negative controls reinstate false claims and must be rejected.

Quantity	Value
Example 16: fresh errors, $\mathbb{E}\varphi$	0.1179
Example 16: mechanism mixture, $\mathbb{E}\varphi$	0.3327
Example 16: linear loss, both	1.0000
Example 16: variance, fresh / mixture	0.900 / 1.476
Theorem 10: $\alpha = 2$, $\beta = 3$, $S = 1$, $\varepsilon = 0$: minimax regret	1.2000

14 Related work and limitations

Risk-based integrity requirements, in which consequence and likelihood set a required failure measure, are standard in functional safety [15]. Our budget is that idea in the gated-computation setting. Choosing a level of automation from the reliability of the automation and the cost of outcomes is an established design question [36], keeping an untrusted model behind trusted monitoring is studied as AI control [37], and expected-loss deferral thresholds go back to the reject option in classification [38].

On the decision side, the worst case over compatible models is the maximum risk of Wald's minimax criterion [17] over a set of models, as in multiple-prior decision theory and partial identification [18, 19]. The twin regret is Le Cam's two-point bound [20] combined with minimax-regret allocation [21, 22]. The limits of statistical validation for very high dependability [23, 24] are the practical content of the second arm. Reviewer conjunction follows the models of coincident failure in diverse software [30, 31, 32], and the errors of language models are empirically correlated [33]. Structural severity is a weighted reach, as in influence spread [34], with risk as probability and consequence [35].

On the probability side, domination by independent variables under conditional bounds is classical [16]. Stochastic and convex orders and their preservation by Bernstein-type operators are standard [10, 7], convex order of mixed binomial counts is known [27], heterogeneous independent counts are convex-below the binomial [28], and the sharp convex bounds from the mean are the Edmundson–Madansky bound and its discrete counterpart [25, 26]; Bernstein convexity is classical [6]; positive association goes back to Harris [8, 9]; infinitely exchangeable sequences are mixtures of independent ones, and the first k terms of an exchangeable binary sequence of length n are within total variation of order k/n of such a mixture, while a whole finite exchangeable vector can be far from every mixture [11]; Ville's inequality, supermartingale Chernoff bounds, and confidence sequences underlie the budget [13, 29, 14].

What we add is the following. Theorem 7 states when closure is mandatory, when it is forbidden, and when no option fits; we are not aware of a prior statement of this two-arm structure. The classical bounds are read as conditions that hold uniformly over the executor's error law, dependence, routing, and mixtures. Structural severity and its properties are stated for gated work, and Proposition 21 reduces the results to Part IV's criterion.

The results compare options through bounds. They do not estimate floors, ceilings, or severity for any real executor, and they do not optimize an adaptive policy. The convex-order results are within conditionally independent processes. Severity factorizes only under the stated condition. The mechanize and gate options interact with maintenance and drift, treated in Parts IV and V and not here. This is a preprint. It has not been peer reviewed. The prior-art search was targeted, not systematic.

15 Conclusion

Severity sets a budget, and the budget, compared with bounds that hold for every compatible error law, decides. Closure is mandatory when it is admissible and every other available option is not; with the budget below every executor floor, an open path with inherited gates is excluded, and only a closure that fits the budget or a gate outside the family remains. Closure is forbidden when an evidence-compatible closed model violates the budget, and when indistinguishable worlds prefer opposite actions, every evidence-based rule pays a worst-case regret proportional to severity. When no option fits, the step needs distinguishing evidence, an exogenous gate, or a redesign. These conditions use only floors, ceilings, certified levels, and compatible twins, so they hold uniformly over error laws, routing, and mixtures of executors within the stated bounds. Persistence matters beyond the mean through convex order, review value is producer-relative unless blind sets are nested, and a martingale budget turns local certificates into a scheduler's lifetime guarantee in expectation, and in tail when damage bounds are pathwise.

Code and data availability

The standard-library Python script `simulate_risk_mechanization.py` reproduces the finite checks and the values of Section 13. It is at https://github.com/ivmat/gated-computation-sim. This paper is archived on Zenodo, DOI 10.5281/zenodo.23324667.

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